

Stage 3E Physics Note: Nonconstant Hidden-Metric Ricci Regression

GL-with-AI Project

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1 Scope

This note summarizes the Stage 3E symbolic regression for nonconstant $\gamma_{ww} = q(x, z)$. It is a derivation artifact only. It does not implement CCZ4 source terms or add GRChombo variables.

2 Metric Ansatz

The regression uses

$$dl^2 = dx^2 + dz^2 + f(x, z)^2 d\Omega_2^2, \quad (1)$$

with

$$f = xP, \quad P = 1 + \lambda x + \mu z, \quad q = P^2. \quad (2)$$

Equivalently,

$$dl^2 = dx^2 + dz^2 + x^2 q(x, z) d\Omega_2^2. \quad (3)$$

Here $q = \gamma_{ww}$ is the hidden cartoon component; the factor x^2 is a coordinate reconstruction factor.

3 Warped-product Reference

For a two-dimensional unit-sphere fiber over the flat base (x, z) , the independent reference formulas are

$$R_{AB} = -2f^{-1} \partial_A \partial_B f, \quad A, B \in \{x, z\}, \quad (4)$$

and

$$R_{\theta\theta} = 1 - |\nabla f|^2 - f \Delta f, \quad (5)$$

$$R_{\phi\phi} = (1 - |\nabla f|^2 - f \Delta f) \sin^2 \theta, \quad (6)$$

where

$$|\nabla f|^2 = f_x^2 + f_z^2, \quad \Delta f = f_{xx} + f_{zz}. \quad (7)$$

The Ricci scalar reference is

$$R = -4 \frac{\Delta f}{f} + 2 \frac{1 - |\nabla f|^2}{f^2}. \quad (8)$$

4 Regression Checks

The SymPy script computes Ricci directly from the metric and asserts agreement with the warped-product reference for R_{xx} , R_{zz} , R_{xz} , $R_{\theta\theta}$, $R_{\phi\phi}$, and R . It also asserts that $\partial_x q$, $\partial_z q$, and $\partial_x \partial_z q$ are not identically zero, verifies the flat special case $\lambda = \mu = 0$, and checks the nontrivial substitution $\lambda = 2, \mu = 3$.

5 Caveats

The Stage 3E script recomputes Ricci using its own copy of symbolic engine helpers such as `christoffel_symbols` and `ricci_tensor`. It therefore directly guards the Ricci engine on a non-constant warped-product case with active q -derivative terms. It does not directly substitute the nonconstant profile into the exact abstract Ricci expressions printed by the Stage 3D script. The Stage 3D expressions are validated transitively because they were produced by the same kind of engine, which is now checked on flat limits, the constant- q_0 cone, and this nonconstant warped-product regression. The residual risk is that a wiring mistake could corrupt the printed or assembled Stage 3D abstract expression while leaving the engine functions intact.

The derivation scripts now duplicate symbolic engine helper functions. This is acceptable short-term, but a future cleanup should factor common helpers into a shared module such as `docs/derivations/cartoon_geometry_common.py`.

For the chosen profile $f = x(1 + \lambda x + \mu z)$, one has $f_{zz} = 0$, so the warped-product formula gives

$$R_{zz} = -2f^{-1}f_{zz} = 0. \quad (9)$$

This does not make the z -derivative sector vacuous: $\partial_z q$ is nonzero, mixed derivative structure is active, $R_{xz} = -2\mu/f$ is generally nonzero, and the angular Ricci components contain μ^2 terms through $|\nabla f|^2$. The narrower limitation is that a bug appearing only in R_{zz} when $f_{zz} \neq 0$ would not be caught by this profile. A future optional refinement could use a profile quadratic in z , making R_{zz} a nonzero target.

6 Limitations

This is still not a full CCZ4 RHS derivation. Off-diagonal g_{xz} , non-flat base metric components, direct substitution into the exact Stage 3D printed abstract Ricci expression, a nonconstant profile with $f_{zz} \neq 0$, conformal variables χ, h_{ww} , regularized small- x limits, C++ translation, unit-test fixtures, and a shared symbolic-geometry helper module remain future work.